

Adversarial Quiz — Answer Key & Rationale

ARI5118 · Non-Maximum Suppression & Detection Evaluation Metrics

Q1. If you run standard NMS with the IoU threshold set to 0.0, what is the effect?

- A. All input boxes are retained, since the threshold is trivial.
- B. Any lower-scoring box that overlaps at all with a higher-scoring box is suppressed.**
- C. Only the single highest-scoring box in the entire image survives, regardless of overlap.
- D. The algorithm degenerates to a no-op.

Correct: B. Suppression fires when $\text{IoU} > 0$, so any positive overlap kills a lower-scoring box; disjoint boxes still survive. A and D invert the threshold direction (treating 0 as "do nothing"). C is the seductive trap: a careful reader assumes "only one survives", forgetting that two non-overlapping boxes have $\text{IoU} = 0$ and both pass.

Q2. Two bounding boxes do not overlap at all. Their IoU is:

- A. Undefined.
- B. Negative.
- C. Exactly 0.**
- D. Equal to 1 minus their distance.

Correct: C. Intersection area is 0 and union area is positive, so $\text{IoU} = 0/\text{union} = 0$ — well defined. A imagines a division-by-zero (it is the intersection that is 0, not the union). B confuses IoU with signed metrics; IoU is bounded in $[0, 1]$. D conflates IoU with GIoU/DIoU , which can be negative for disjoint boxes — plain IoU cannot.

Q3. Boxes A and B perfectly overlap ($\text{IoU} = 1.0$). A has score 0.9, B has score 0.7. Standard NMS with IoU threshold 0.5 outputs:

- A. Both A and B, since they are identical.
- B. Only A.**
- C. Neither, since duplicates are discarded.
- D. An average of A and B.

Correct: B. $\text{IoU} = 1.0 > 0.5$, so the lower-scoring duplicate B is suppressed and A survives. A misreads "identical" as a reason to keep both identical boxes are exactly the strongest case for suppression. C imagines a symmetric deletion rule; NMS never removes the top-scoring box. D confuses NMS with Weighted Box Fusion: standard NMS never averages coordinates.

Q4. Which of the following is NOT a required input to the standard NMS algorithm?

- A. A set of candidate bounding boxes.
- B. Per-box confidence scores.
- C. An IoU threshold.
- D. Ground-truth boxes.**

Correct: D. NMS is a pure post-processing pass over predictions; ground truth is only consulted at evaluation time. A, B, C are all genuinely required: the boxes themselves, the scores that decide which one to keep, and the IoU threshold that defines "overlapping". Picking D wrong means imagining NMS as "evaluation-aware", a classic source of train/inference data leakage in novice implementations.

Q5. A detector outputs two boxes of the same class: Box A (confidence 0.40) and Box B (confidence 0.70), with $\text{IoU}(A, B) = 0.50$. Standard NMS is then run with a confidence threshold of 0.50. Which box is suppressed?

- A. Box A, because it has the lower confidence and $\text{IoU}(A, B) \geq 0.50$.
- B. Box B, because its confidence equals the threshold and one box must be removed when two overlap.
- C. Both boxes survive, because their IoU is exactly equal to the (assumed) overlap threshold of 0.50, which is not strictly greater than it.
- D. The question cannot be answered as stated — NMS is gated by an IoU threshold N_t , not by the confidence threshold, and no N_t has been specified.**

Correct: D. The confidence threshold pre-filters which boxes enter NMS; the IoU threshold N_t decides suppression inside NMS. The question gives only the former, so suppression is undetermined. A treats 0.50 as if it were the IoU threshold — the most common mistake. B invents a "one must die" rule; NMS keeps the highest scorer unconditionally. C smuggles in an unstated $N_t = 0.50$, constructing an answer from assumed values is exactly the failure mode the question targets.

Q6. A team replaces standard NMS with Soft-NMS. Their precision collapses: the system now emits tens of thousands of low-confidence boxes per image. Which parameter is the decisive filter that determines what ends up in the final output set?

- A. The IoU overlap threshold N_t , since Soft-NMS uses the same overlap criterion as standard NMS.
- B. The σ parameter in the Gaussian decay, since it controls how aggressively overlapping boxes are suppressed.
- C. The final confidence-score threshold applied after rescaling, because Soft-NMS only decays scores, it never removes boxes itself.**
- D. The detection-evaluation IoU threshold O_t , since it defines which surviving boxes count as true positives.

Correct: C. Soft-NMS decays scores rather than deleting boxes, so the only thing that prunes the output is the post-rescaling score threshold, forgetting it is the canonical Soft-NMS bug. A conflates "what the algorithm reads" with "what gates the output"; IoU only feeds the decay function. B controls decay shape but redistributes scores within the ranked list; a box decayed to 0.0001 still survives without a final threshold. D acts downstream O_t judges output but cannot reduce its size.

Q7. Soft-NMS yields the largest improvement over standard NMS in which setting?

- A. Sparse scenes with one isolated object.
- B. Scenes with multiple legitimately overlapping objects of the same class.**
- C. Single-class detection on uncluttered backgrounds.
- D. Whenever the detector outputs perfectly localised boxes.

Correct: B. Crowded same-class scenes are exactly where standard NMS over-suppresses real detections the Soft-NMS paper's motivating example. A and C describe sparse/uncluttered settings where NMS and Soft-NMS behave nearly identically. D confuses the role of Soft-NMS: it improves which boxes survive, not how tightly they localise localisation is set by bounding-box regression, not by post-processing.

Q8. With strong class imbalance, mAP:

- A. Is biased toward majority classes because it weights detections.
- B. Gives equal weight to each class — minority classes count just as much per-class as majority ones.**
- C. Cannot be computed.
- D. Reduces to balanced accuracy.

Correct: B. Standard mAP is per-class AP first, then an unweighted mean over classes: a 5-instance class counts equally with a 5,000-instance class. A is tempting because rare classes have fewer detections, but averaging happens after per-class AP. C invents a "metric breaks under imbalance" rule; mAP is well-defined regardless. D borrows a classification concept; balanced accuracy is not the same metric as mAP.

Q9. A detector outputs the same set of detections as another but with all confidence scores multiplied by 0.1 (a miscalibrated but order-preserving rescaling). Compared to the original, its AP will be:

- A. Strictly lower.
- B. Identical — AP depends on the ranking of detections, not absolute score values.**
- C. Zero.
- D. Lower at high IoU thresholds only.

Correct: B. AP is computed by sweeping the confidence threshold and accumulating TPs/FPs in rank order; any monotone-preserving transform of scores leaves the rank order and thus AP unchanged. A is the canonical calibration confusion. C imagines that low scores zero the metric, but AP cares about rank, not magnitude. D mistakenly couples rescaling to localisation strictness; IoU thresholds and score scaling are independent.

Q10. In COCO evaluation, a second predicted box that overlaps an already-matched ground-truth box is counted as:

- A. A True Positive.
- B. A False Positive.**
- C. Ignored (does not affect the metric).
- D. A True Negative.

Correct: B. Each ground truth can be matched at most once; subsequent overlapping predictions become FPs. A would let a detector game AP by replicating the same prediction. C ignores the duplicate, but duplicates do damage precision and are counted. D imports a classification concept: True Negatives are essentially undefined in detection because the negative class, every empty patch of pixels, is not enumerable.